Spectral Theorem

25. **(T/F)** An $n \times n$ matrix that is orthogonally diagonalizable must be symmetric.

26. **(T/F)** There are symmetric matrices that are not orthogonally diagonalizable.

27. **(T/F)** An orthogonal matrix is orthogonally diagonalizable.

28. **(T/F)** If $B = PDP^T$, where $P^T = P^{-1}$ and D is a diagonal matrix, then B is a symmetric matrix.

29. **(T/F)** For a nonzero $\mathbf{v}$ in $\mathbb{R}^n$, the matrix $\mathbf{v}\mathbf{v}^T$ is called a projection matrix.

30. **(T/F)** If $A^T = A$ and if vectors $\mathbf{u}$ and $\mathbf{v}$ satisfy $A\mathbf{u} = 3\mathbf{u}$ and $A\mathbf{v} = 4\mathbf{v}$, then $\mathbf{u} \cdot \mathbf{v} = 0$.

31. **(T/F)** An $n \times n$ symmetric matrix has n distinct real eigenvalues.

32. **(T/F)** The dimension of an eigenspace of a symmetric matrix is sometimes less than the multiplicity of the corresponding eigenvalue.

33. Show that if A is an $n \times n$ symmetric matrix, then $(A\mathbf{x}) \cdot \mathbf{y} = \mathbf{x} \cdot (A\mathbf{y})$ for all $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^n$.

34. Suppose A is a symmetric $n \times n$ matrix and B is any $n \times m$ matrix. Show that B^TAB, B^TB, and BB^T are symmetric matrices.

35. Suppose A is invertible and orthogonally diagonalizable. Explain why A^{-1} is also orthogonally diagonalizable.

36. Suppose A and B are both orthogonally diagonalizable and $AB = BA$. Explain why AB is also orthogonally diagonalizable.

37. Let $A = PDP^{-1}$, where P is orthogonal and D is diagonal, and let λ be an eigenvalue of A of multiplicity k. Then λ appears k times on the diagonal of D. Explain why the dimension of the eigenspace for λ is k.

38. Suppose $A = PRP^{-1}$, where P is orthogonal and R is upper triangular. Show that if A is symmetric, then R is symmetric and hence is actually a diagonal matrix.

Singular Value Decomposition

Find an SVD of each matrix in Exercises 5–12. [*Hint:* In Exercise 11, one choice for U is $\begin{bmatrix} -1/3 & 2/3 & 2/3 \\ 2/3 & -1/3 & 2/3 \\ 2/3 & 2/3 & -1/3 \end{bmatrix}$. In Exercise 12, one column of U can be $\begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$.

5. $\begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$

6. $\begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix}$

7. $\begin{bmatrix} 2 & -1 \\ 2 & 2 \end{bmatrix}$

8. $\begin{bmatrix} 4 & 6 \\ 0 & 4 \end{bmatrix}$

9. $\begin{bmatrix} 3 & -3 \\ 0 & 0 \\ 1 & 1 \end{bmatrix}$

10. $\begin{bmatrix} 7 & 1 \\ 5 & 5 \\ 0 & 0 \end{bmatrix}$

11. $\begin{bmatrix} -3 & 1 \\ 6 & -2 \\ 6 & -2 \end{bmatrix}$

12. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$

Find the SVD of $A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$ [*Hint:* Work with A^T.]

In Exercise 7, find a unit vector $\mathbf{x}$ at which $A\mathbf{x}$ has maximum length.

Suppose the factorization below is an SVD of a matrix A, with the entries in U and V rounded to two decimal places.

$$A = \begin{bmatrix} .40 & -.78 & .47 \\ .37 & -.33 & -.87 \\ -.84 & -.52 & -.16 \end{bmatrix} \begin{bmatrix} 7.10 & 0 & 0 \\ 0 & 3.10 & 0 \\ 0 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} .30 & -.51 & -.81 \\ .76 & .64 & -.12 \\ .58 & -.58 & .58 \end{bmatrix}$$

a. What is the rank of A?

b. Use this decomposition of A, with no calculations, to write a basis for Col A and a basis for Nul A. [*Hint:* First write the columns of V.]

In Exercises 17–24, A is an $m \times n$ matrix with a singular value decomposition $A = U\Sigma V^T$, where U is an $m \times m$ orthogonal matrix, Σ is an $m \times n$ "diagonal" matrix with r positive entries and no negative entries, and V is an $n \times n$ orthogonal matrix. Justify each answer.

17. Show that if A is square, then $|\det A|$ is the product of the singular values of A.

18. Suppose A is square and invertible. Find a singular value decomposition of A^{-1}.

Image Compression

Exercise 10.1. Consider the following image:

(a) Enter this image into a 4×4 matrix A. Assume the shades you see are only 0, 0.5 and 1.

(b) Find the SVD for A.

(c) Simplify the matrix preserving 3, 2 and then 1 singular value. Draw the best image representation you can of each matrix.

Exercise 10.3. Find a square image on the internet of reasonable size; 200×200 or thereabouts is good.

(a) Load this into the Matlab matrix A, convert to grayscale if necessary and scale the values between 0 and 1.

(b) Find the SVD for A.

(c) How many singular values are there?

(d) Simplify the image by preserving only 75% of the singular values. What is the percentage of data quality preserved? Print the result.

(e) Simplify the image by preserving only 50% of the singular values. What is the percentage of data quality preserved? Print the result.

(f) Simplify the image by preserving only 25% of the singular values. What is the percentage of data quality preserved? Print the result.

(g) What is the minimum number of singular values that must be preserved in order to preserve 99.9% of the data quality? What level of data compression would this achieve? Simplify the image accordingly and print the result.

Exercise 10.6. Suppose we simplify an $m \times n$ (not necessarily square) image using k singular values. What is the resulting data compression ratio? What criteria on k would make this worth doing?

Exercise 2.2. Which pairs of expressions for quantum states represent the same state? For those pairs that represent different states, describe a measurement for which the probabilities of the two outcomes differ for the two states and give these probabilities.

a. $|0\rangle$ and $-|0\rangle$

b. $|1\rangle$ and $\mathbf{i}|1\rangle$

c. $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(-|0\rangle + \mathbf{i}|1\rangle)$

d. $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

e. $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ and $\frac{1}{\sqrt{2}}(|1\rangle - |0\rangle)$

f. $\frac{1}{\sqrt{2}}(|0\rangle + \mathbf{i}|1\rangle)$ and $\frac{1}{\sqrt{2}}(\mathbf{i}|1\rangle - |0\rangle)$

g. $\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|0\rangle$

h. $\frac{1}{\sqrt{2}}(|\mathbf{i}\rangle - |-\mathbf{i}\rangle)$ and $|1\rangle$

i. $\frac{1}{\sqrt{2}}(|\mathbf{i}\rangle + |-\mathbf{i}\rangle)$ and $\frac{1}{\sqrt{2}}(|-\rangle + |+\rangle)$

j. $\frac{1}{\sqrt{2}}\left(|0\rangle + e^{\mathbf{i}\pi/4}|1\rangle\right)$ and $\frac{1}{\sqrt{2}}\left(e^{-\mathbf{i}\pi/4}|0\rangle + |1\rangle\right)$

Exercise 2.3. Which states are superpositions with respect to the standard basis, and which are not? For each state that is a superposition, give a basis with respect to which it is not a superposition.

a. $|+\rangle$

b. $\frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$

c. $\frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$

d. $\frac{\sqrt{3}}{2}|+\rangle - \frac{1}{2}|-\rangle)$

e. $\frac{1}{\sqrt{2}}(|\mathbf{i}\rangle - |-\mathbf{i}\rangle)$

f. $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

Exercise 2.4. Which of the states in 2.3 are superpositions with respect to the Hadamard basis, and which are not?

Exercise 2.6. For each pair consisting of a state and a measurement basis, describe the possible measurement outcomes and give the probability for each outcome.

a. $\frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle, \{|0\rangle, |1\rangle\}$

b. $\frac{\sqrt{3}}{2}|1\rangle - \frac{1}{2}|0\rangle, \{|0\rangle, |1\rangle\}$

c. $|-\mathbf{i}\rangle, \{|0\rangle, |1\rangle\}$

d. $|0\rangle, \{|+\rangle, |-\rangle\}$

e. $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \{|\mathbf{i}\rangle, |-\mathbf{i}\rangle\}$

f. $|1\rangle, \{|\mathbf{i}\rangle, |-\mathbf{i}\rangle\}$

g. $|+\rangle, \{\frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle, \frac{\sqrt{3}}{2}|0\rangle - \frac{1}{2}|1\rangle\}$